Dominik Winterer

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ACADEMIC EXPERIENCE

- Lecturer in Cyber Security, Department of Computer Science, University of Manchester, 09/2025–present
- PostDoc, Department of Computer Science, ETH Zurich, 11/2024–05/2025
- Ph.D., Computer Science, ETH Zurich, 03/2019–10/2024
- MSc., Computer Science, University of Freiburg, 2018
- BSc., Computer Science, University of Freiburg, 2011

INDUSTRIAL EXPERIENCE

- Applied Scientist Intern, AWS Automated Reasoning Group, 10/2023 – 01/2024
- Research Intern, IBM Research Yorktown Heights, 06/2017—09/2017

SELECTED PUBLICATIONS

- Dominik Winterer, Zhendong Su. Validating SMT Solvers for Correctness and Performance via Grammar-based Enumeration *In Proceedings of SPLASH/OOPSLA 2024*.
- Mauro Bringolf, Dominik Winterer, Zhendong Su. Finding and Understanding Incompleteness Bugs in SMT Solvers. *In Proceedings of ASE 2022*.
- Jiwon Park, Dominik Winterer, Chengyu Zhang, and Zhendong Su. Generative Type-Aware Mutation for Testing SMT Solvers. *In Proceedings of SPLASH/OOPSLA 2021*.
- Dominik Winterer, Chengyu Zhang, and Zhendong Su. On the Unusual Effectiveness of Operator Mutations for Testing SMT Solvers. *In Proceedings of SPLASH/OOPSLA 2020*.
- Dominik Winterer, Chengyu Zhang, and Zhendong Su. Validating SMT Solvers via Semantic Fusion. *In Proceedings of PLDI 2020*.
  PLDI Distinguished Paper Award
  Invited to ACM TOPLAS Special Issue

AWARDS & GRANTS

- Selected participant for 9th Heidelberg Laureate Forum, 2022
- **Amazon Research Award (80,000 USD, Co-I), 2021**
- Google Open Source Peer Bonus, 2021
- **ACM Distinguished Paper Award at PLDI '20**
- IJCAI '16 travel grant (1,000 USD)
- Grant by "Freunde der Universität Freiburg" in 2016 (700 USD)

IMPACT

- Tools: yinyang (SMT solver testing framework, 198 stars), ET (exhaustive testing tool, 22 stars)
- Bug findings: 1,800+ bugs in the SMT solvers Z3 and CVC4/cvc5
- Service: SMT-COMP organizer (2025–present), PC Member: PLDI '26, Euro S&P '26, CAV '25, ASE '25, ICSE 2024 (Demo track), AAAI '20, AAAI '21, SYNASC '22, MET '23

INVITED TALKS

- Formal Methods Engineering: Toward Reliable Software Engineering
  Cornell & UT Austin SE seminar (virtual), Dec 2025
  MPI-SP Sparkling Series, Apr 2025
  USI Lugano Software Seminar Series, Feb 2025
  UC Riverside (faculty interview), Jan 2025
- Solidifying SMT Solvers through Automated Testing Techniques
  Harvard PL Seminar, Mar 2024
  UPenn, PLClub. Feb 2024
  UC Berekely, Programming Systems Seminar, Jan 2024
- Directed Testing of a String Solver, Feb 2024
  Amazon Webservices, Feb 2024
- Finding 1,700+ Bugs in the SMT Solvers Z3 and CVC5
  UCSC LSD Seminar (virtual), Feb 2023
  USC Software Seminar (virtual), Nov 2022
  KIT Research Seminar Formal Methods, Sep 2022
  Paderborn University, Jun 2022
  CEA List at Paris Saclay (virtual), Feb 2021

SERVICE

- Co-organizer of the SMT-COMP (since 2025)
- Chair of the ASE '26 Student Research Competition
- ISSTA '24 Artifact Evaluation Co-Chair
- Program Committee Member
  - ICSE '27, FSE '27, OOPSLA '27, FUZZING '25, ASE '25, CAV '25, Euro S&P '26, PLDI '26
  - ICSE '24 Demonstration track, OOPSLA '24 SRC
  - AAAI '20, AAAI '21
- Journal Reviewer: TSE '21, JAIR '17, TOSEM '24
- Artifact Evaluation Committee Member: PLDI '22, POPL '21, ISSTA '21

SUPERVISION & MENTORING

At University of Manchester

Lu Maltsis, Ph.D. Student, Jan 2026 - present

Nikita Kroshko, BSc. Student intern, Jan 2026 - Mar 2025

Joshua Bode, BSc. Student intern, Jan 2026 - Mar 2025

Andrei Zhukov, MSc. Thesis, Aug 2024 - Mar 2025

At ETH Zurich

Thea Kjeldsmark, Undergrad from UC Irvine, ETH Summer Research Fellow, Jun 2024 - 2024
POPL '25 Student Research Competition Participant
→ Ph.D. Student at UC Irvine

Pasquale Jordan, MSc. Thesis, Aug 2024 - Mar 2025

Lu Maltsis, BSc. Thesis, Jan 2024 - Aug 2024
→ Ph.D. Student at University of Manchester

Marc Carigiet, BSc. Thesis, Sep 2023 - Dec 2023

Vince Szabó, Undergrad from TU Delft, ETH Summer Research Fellow, Jun 2023 - Sep 2023

POPL '24 Student Research Competition Participant
Arxiv '24

Altin Alickaj, MSc. practical work, Sep 2022 - Jan 2023
→ Research Scientist at Oracle Labs, Zurich

Michael Mörschell, BSc. Thesis, Mar 2023 - Aug 2023
→ MSc. ETH Zurich

Jiwon Park, Research Intern, Jan 2021 - Mar 2021 & Sep 2021 - Dec 2021
2nd place at FSE '21 Student Research Competition
Co-first Author of OOPSLA '21 paper
→ Ph.D. Student at UC Berkeley

Mauro Bringolf, MSc. Thesis, Feb 2021 - Aug 2021
First author of ASE '22 paper

Dylan J Wolff, MSc. Thesis, Jan 2020 - Jul 2020
POPL '21 Student Research Competition Participant
→ Ph.D. Student at National University of Singapore

Roxana Mafteiu-Scai, Undergrad from HAW Hamburg, Jul 2020 - Sep 2020
→ MSc. Student at University of Vienna

SIGPLAN-M Longterm Mentoring (Oct 2021 - present)

Anirudh Basu, Undergrad from UC Davis, advice on applying for Ph.D. positions
Mayank Keoliya, Undergrad from NUS, advice on applying for Ph.D. positions
→ Ph.D. Student at University of Pennsylvania
Theodore Leebrant, Undergrad from NUS, career advice

TEACHING

At University of Manchester

- Software Security (80-100 students, Lecturer), Spring 2026
- Software Engineering (300+ students, Lecturer), Fall 2025

At ETH Zurich

- **Head Teaching Assistant**
  - Compiler Design (200+ students, 9 TAs), Fall 2021
  - Software Engineering (100+ students, 5 TAs, Guest Lecturer), Spring 2022
- **Teaching Assistant**
  - Compiler Design, Fall 2019, 2020, 2022
  - Data Modeling and Databases, Spring 2020, 2023
  - Formal Methods and Functional Programming, Spring 2022
  - Algorithms & Data Structures, Spring 2024
  - Automated Software Testing (Mentor + Guest Lecturer), Spring 2021, 2022, 2023, 2024
  - Seminar: Research Topics in Software Engineering, Spring 2020, 2021, 2022, Fall 2020